



WHO HIV Data Analysis Dashboard

Healthcare Data Analytics Project

Created by : Onye Nnenna

Tools Used: Power BI, DAX, Power Query

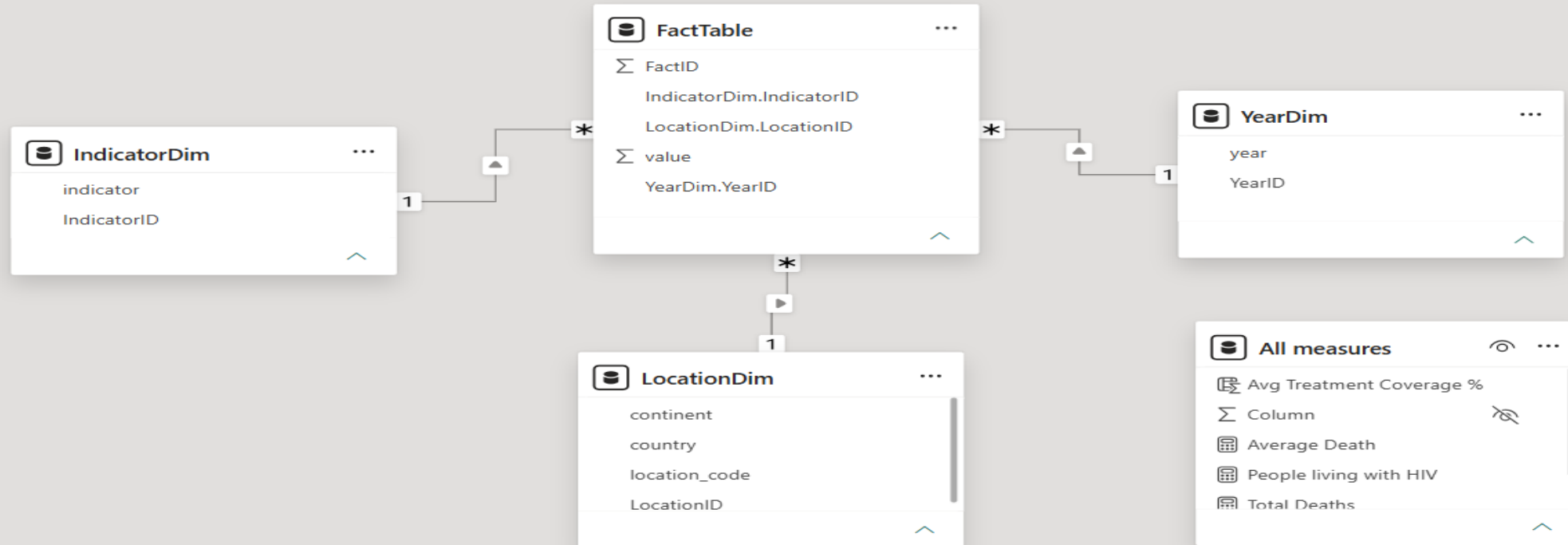


Project Overview

This project analyzes global HIV trends using WHO datasets to uncover insights into deaths, treatment coverage, and survival outcomes globally.

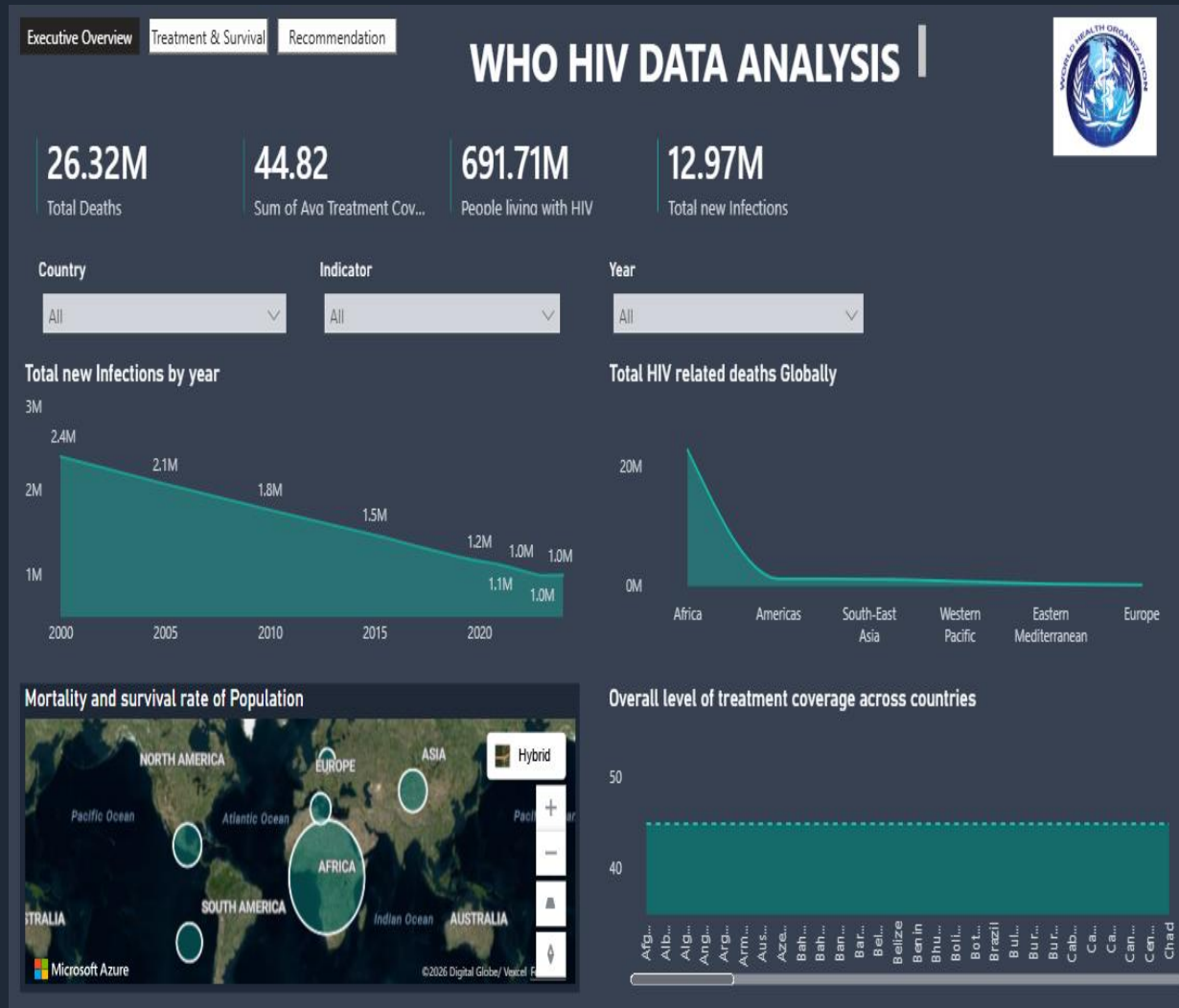
The dashboard was developed using Power BI, DAX, and Power query to support healthcare decision making through data driven insights.

Data Modeling



The dataset was transformed into a clean star schema model consisting of one Fact table Multiple dimension tables including Year Dim , Indicator Dim , Location Dim. Dax measures were created to support KPI calculations and time intelligence analysis.

Executive Overview Dashboard



Insights

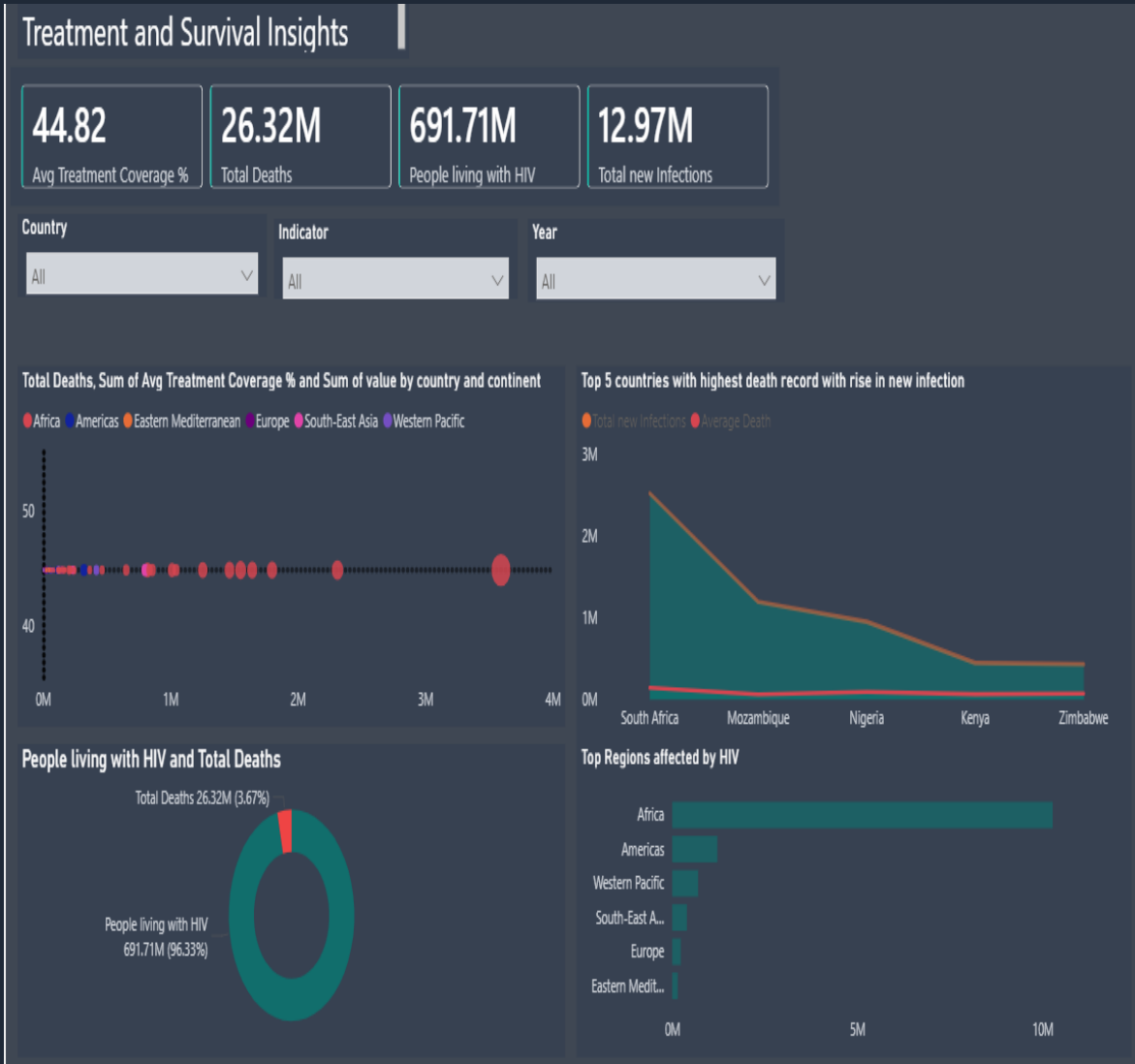
HIV Infections declined over time globally.

South Africa recorded the highest level of new infections

HIV related deaths shows a significant low trend generally but the trend is still high in some parts of Africa as seen in the dashboard.

Treatment coverage improved significantly globally leading to increased survival outcomes and decreased mortality .

Treatment and Survival Insights



Insights

Countries in Africa are the most affected with new cases of HIV.

Africa is the region associated with the most HIV related deaths in total population.

Africa is the top region affected by HIV followed by America.

Data shows that globally there is an equal average treatment coverage.

Recommendation

INSIGHTS AND RECOMMENDATIONS

- HIV deaths have declined over time globally
- African countries tends to have the highest rate of people living with HIV
- ART coverage shows correlation with improved survival outcome globally
- People living with new infection are decreasing but remain high in most African countries.
- Increase ART coverage in African countries where there is increase in number of new cases.
- The analysis reveals a steady decline in new HIV infections globally from 2000 onwards, indicating progress in prevention and treatment initiatives.

RECOMMENDATIONS

- Increase ART accessibility in high risk countries
- Strengthen awareness and prevention programs in countries like Nigeria and South Africa and Africa at large.
- Improve healthcare funding in countries with high rate of people living with HIV. creation of more testing sites and ART medication access and steady supply.

Data source: WHO Global HIV Dataset
Created By: Onye Nnenna

Insights

Increase ART accessibility in high risk countries.

Expand HIV awareness and prevention programs across Africa.

Increase funding to improve testing facilities are expanded into rural areas and made easily accessible .

Strengthen treatment initiative globally.

Ensure constant and monitored viral load tests.

Conclusion

This project demonstrates how healthcare data analytics can support informed public health decision and improve HIV intervention strategies globally.

The dashboard highlights regions with increasing HIV infections and identifies countries with high mortality. infection and new infection rates.

The analysis has revealed that that improved treatment coverage and ART accessibility have contributed significantly to reduction in HIV infection related deaths since 2000-2024.

However, despite the global decline in mortality , new infections , new infections remain high in several regions, indicating the need for continuous and increased support and investment in prevention, awareness, testing, treatment and accessibility globally.

THANK YOU.

Questions?